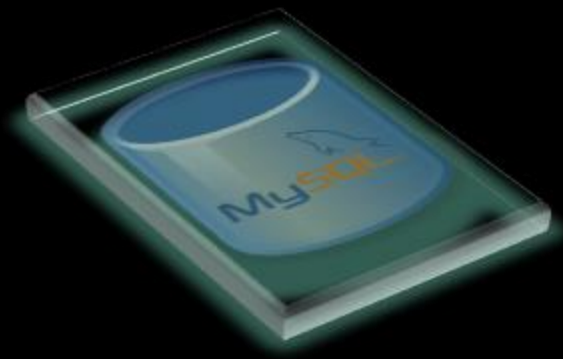


DATA EXPLORATION & BUSINESS INSIGHTS WITH **SQL** (NORTHWIND DATABASE)





PROJECT INTRODUCTION



Database

Northwind database
simulates sales and
customer relations.

1

SQL IDE

DBeaver is used to
explore table
relationships.

2

Business Questions

SQL queries answer
key business
questions.

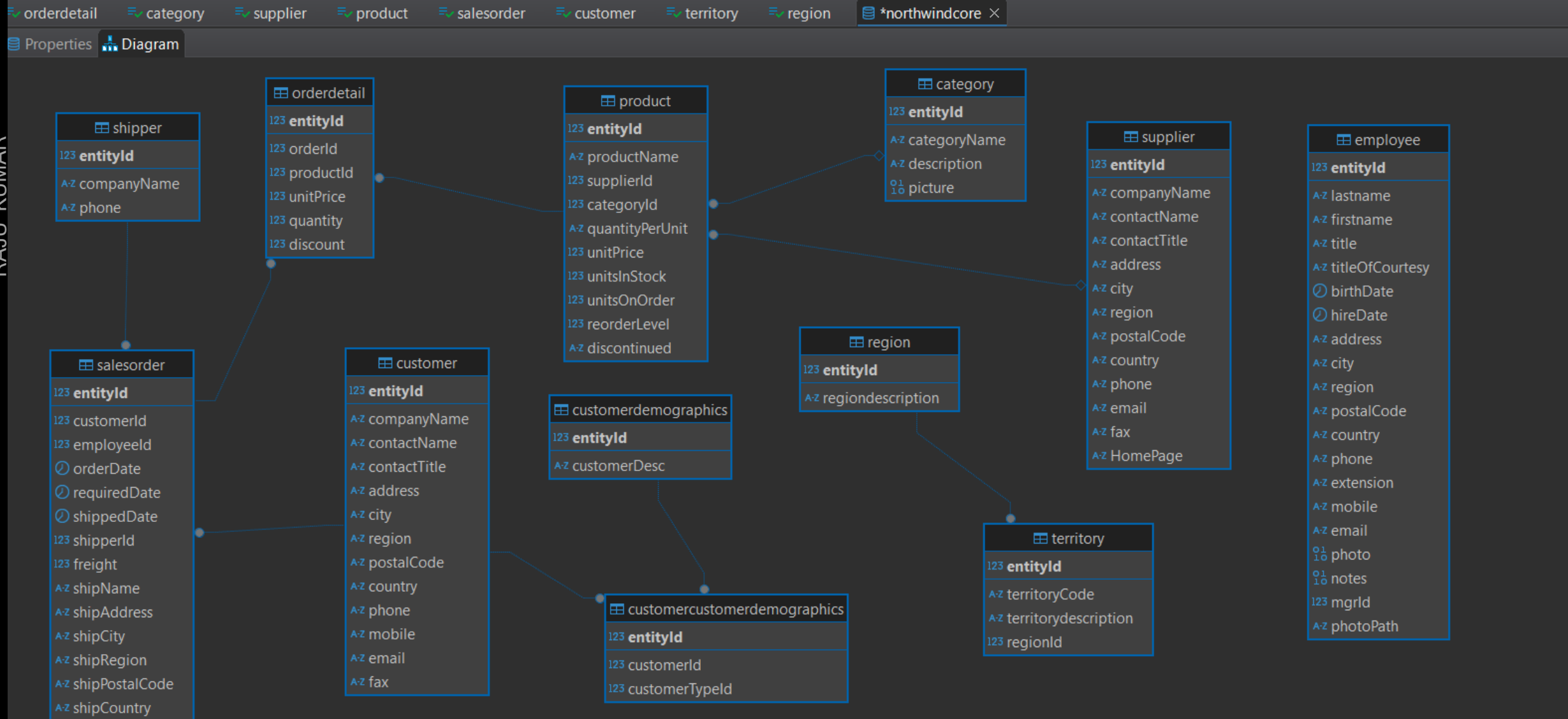
3

Database Schema Overview

- The schema includes core entities such as:
 - salesorder, orderdetail, customer, product, employee, supplier, etc.
- Tables are linked via **primary and foreign keys** to maintain data integrity.
- This logical structure allows us to perform **complex joins**, aggregations, and filters to generate business reports.
- **Purpose:** Data querying, filtering, aggregation, and visualization (if required)



RELATIONAL DATABASE STRUCTURE



-- WHO ARE THE TOP 10 CUSTOMERS BASED ON TOTAL PURCHASES?

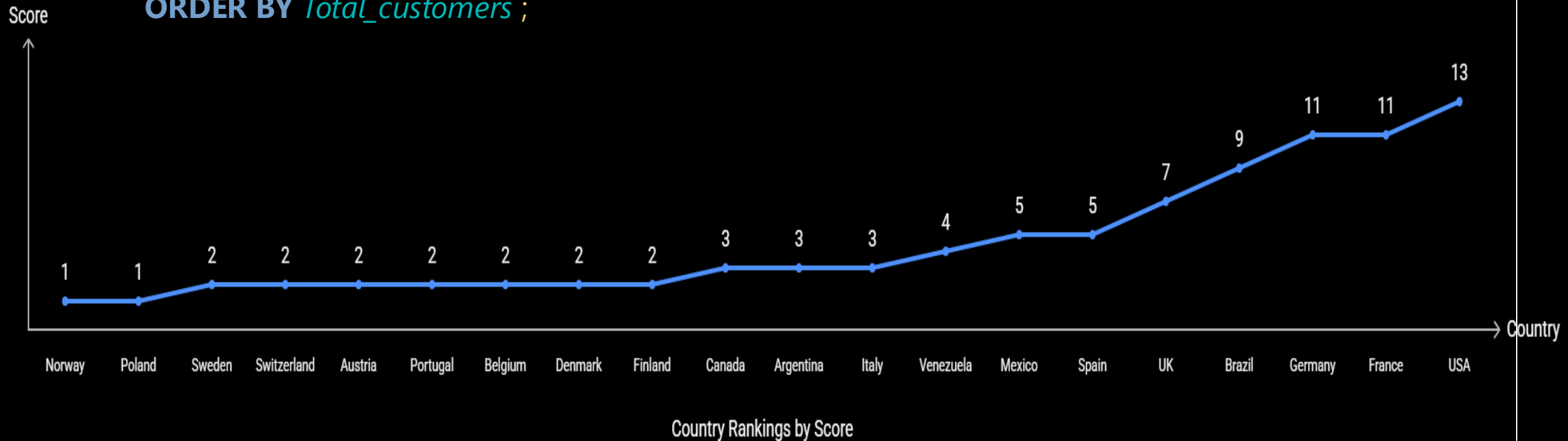
```
SELECT C.CONTACTNAME , SUM(O.UNITPRICE * O.QUANTITY) as
total_purchases
from
NORTHWINDCORE.CUSTOMER C
join
NORTHWINDCORE.SALESORDER S on C.ENTITYID =
S.CUSTOMERID
join
NORTHWINDCORE.ORDERDETAIL O on S.ENTITYID = O.ORDERID
GROUP BY C.ENTITYID
ORDER BY total_purchases DESC
limit 10;
```

Top 10 Customers by Purchases

	 Total Purchases
Veronesi, Giorgio	117483.39
Navarro, Tomás	115673.39
Kane, John	113236.68
Crăciun, Ovidiu V.	57317.39
Moore, Michael	52245.90
Cohen, Shy	34101.15
San Juan, Patricia	32555.55
Taylor, Maurice	32203.90
Song, Lolan	31745.75
Misieć, Anna	30226.10

-- WHAT IS THE TOTAL NUMBER OF CUSTOMERS IN EACH COUNTRY?

```
SELECT COUNTRY , Count(*) as Total_customers from  
NORTHWINDCORE.CUSTOMER C  
group by COUNTRY  
ORDER BY Total_customers ;
```



-- WHAT IS THE MOST COMMON CUSTOMER TYPE (CUSTOMER TYPE ID)?

```
SELECT CUSTOMERID , COUNT(*) as  
customer_count  
from  
NORTHWINDCORE.SALESORDER S  
group by CUSTOMERID  
order by customer_count desc ;
```

-- WHICH CUSTOMERS HAVE THE HIGHEST DISCOUNT APPLIED TO THEIR ORDERS?

```
SELECT C.CONTACTNAME ,  
Max(O.DISCOUNT) as  
highest_discount  
from  
NORTHWINDCORE.ORDERDETAIL  
O  
join  
NORTHWINDCORE.CUSTOMER C  
on C.ENTITYID = O.ENTITYID  
group by C.CONTACTNAME  
ORDER by highest_discount desc  
;
```

```
with DiscountRank as (  
select C.CONTACTNAME , O.DISCOUNT ,  
DENSE_RANK () over( order by O.DISCOUNT desc ) as  
ranke  
from NORTHWINDCORE.ORDERDETAIL O  
join NORTHWINDCORE.CUSTOMER C  
on O.ENTITYID = C.ENTITYID  
)  
select * from DiscountRank  
order by DISCOUNT desc ;
```

-- WHICH PRODUCTS HAVE THE HIGHEST SALES?

SELECT

P.PRODUCTNAME,

SUM(*O*.QUANTITY * *P*.UNITPRICE) **AS** TOTAL_SALES

FROM NORTHWINDCORE.PRODUCT *P*

JOIN NORTHWINDCORE.ORDERDETAIL *O*

ON *P*.ENTITYID = *O*.PRODUCTID

GROUP BY *P*.PRODUCTNAME

ORDER BY TOTAL_SALES desc

limit 10 ;

RAJU KUMAR

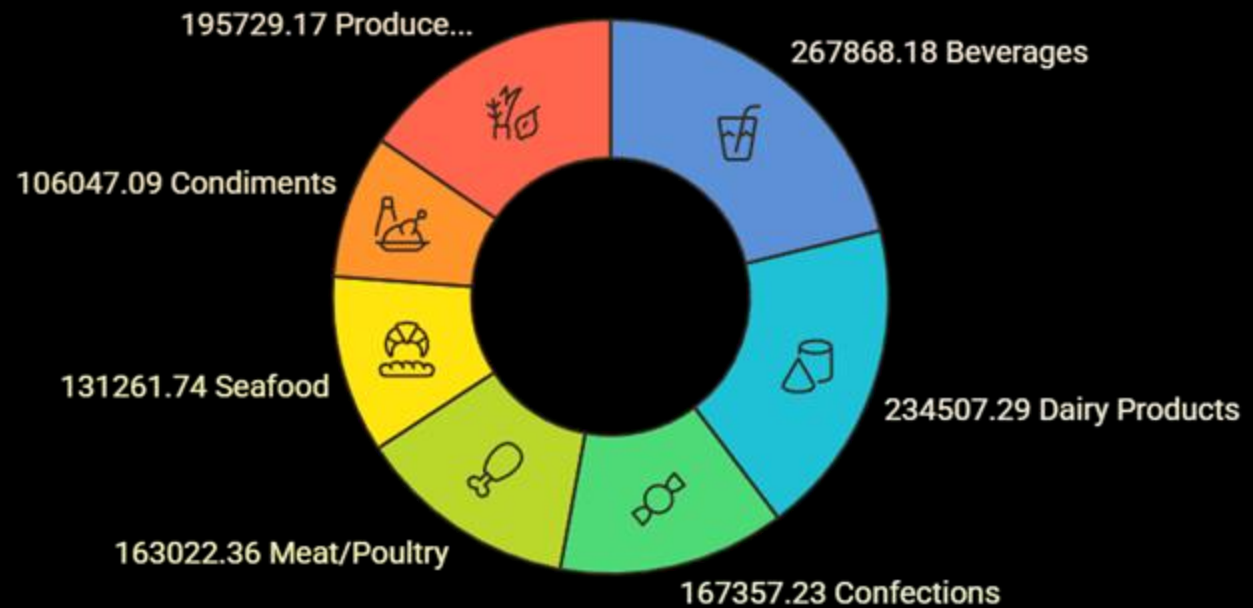


Product Values

-- WHAT IS THE TOTAL REVENUE GENERATED BY EACH PRODUCT CATEGORY?

```
select C.CATEGORYNAME ,  
SUM(O.UNITPRICE * O.QUANTITY  
* (1-O.discount) ) as total_revenue  
from  
NORTHWINDCORE.CATEGORY C  
join  
NORTHWINDCORE.PRODUCT P on  
P.CATEGORYID = C.ENTITYID  
join  
NORTHWINDCORE.ORDERDETAIL  
O on O.PRODUCTID = p.ENTITYID  
group by c.CATEGORYNAME  
order by total_revenue desc ;
```

Distribution of Food and Beverage Categories (units)



Made with Napkin

-- WHICH PRODUCT HAS THE LOWEST STOCK AVAILABLE?

```
select PRODUCTNAME  
,UNITSINSTOCK from  
NORTHWINDCORE.PRODUCT P  
order by UNITSINSTOCK ;
```

-- WHICH SUPPLIER PROVIDES THE MOST PRODUCTS?

```
select companyName , COUNT(*) as ProductSupply from  
NORTHWINDCORE.SUPPLIER S  
join  
NORTHWINDCORE.PRODUCT P  
on P.SUPPLIERID = S.ENTITYID  
GROUP by S.companyName  
order by ProductSupply desc  
limit 1 ;
```

-- WHAT IS THE REORDER LEVEL FOR EACH PRODUCT, AND WHICH ONES NEED RESTOCKING?

```
SELECT productName,  
UNITSINSTOCK, reorderLevel  
  
FROM  
NORTHWINDCORE.PRODUCT  
  
WHERE UNITSINSTOCK <=  
reorderLevel  
  
ORDER BY UNITSINSTOCK;
```

Top Product Performers



-- WHICH EMPLOYEES HAVE PROCESSED THE MOST ORDERS?

```
SELECT CONCAT(E.FIRSTNAME, ' ',  
E.LASTNAME) AS EmployeeName ,  
EMPLOYEEID , count(*) as  
processedOrder  
  
from  
NORTHWINDCORE.SALESORDER S  
  
join NORTHWINDCORE.EMPLOYEE  
E on  
  
E.ENTITYID = S.EMPLOYEEID  
  
group by EMPLOYEEID  
  
order by processedOrder desc  
  
limit 1 ;
```

-- WHAT IS THE AVERAGE NUMBER OF ORDERS HANDLED PER EMPLOYEE?

```
WITH EmployeeOrders AS (  
  SELECT EMPLOYEEID, COUNT(*) AS OrderCount  
  FROM NORTHWINDCORE.SALESORDER  
  GROUP BY EMPLOYEEID  
)  
SELECT AVG(OrderCount) AS AvgOrdersPerEmployee FROM  
EmployeeOrders;
```

92.2222

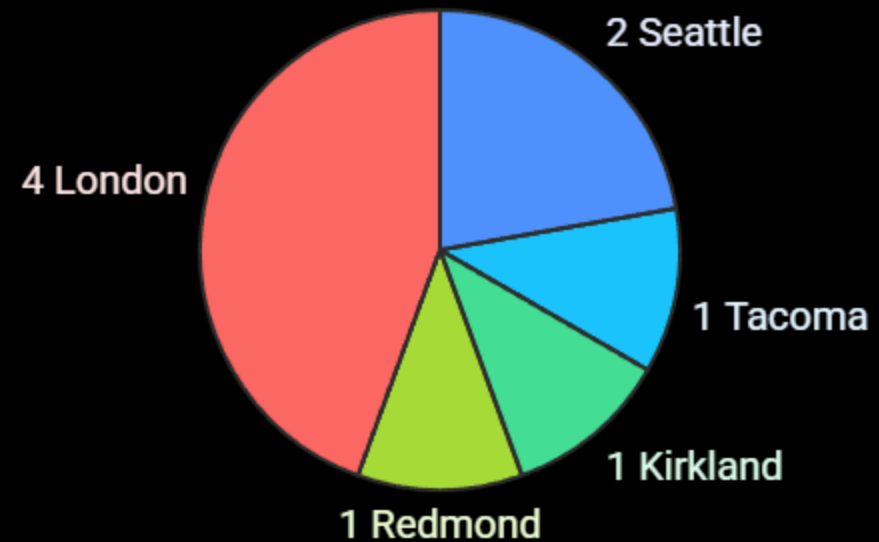
-- WHICH EMPLOYEE HAS BEEN WITH THE COMPANY THE LONGEST (BASED ON HIRE DATE)?

```
SELECT *, ROUND(DATEDIFF(NOW(), HIREDATE) / 365 , 0 ) AS  
YearsSinceHired  
FROM NORTHWINDCORE.EMPLOYEE  
order by YearsSinceHired desc  
limit 1 ;
```

-- WHAT IS THE DISTRIBUTION OF EMPLOYEES BY REGION OR CITY?

```
select  
  CITY ,COUNT(*) as distributionOfEmployees  
from NORTHWINDCORE.EMPLOYEE E  
group by CITY ;
```

**Distribution of Occurrences by Location
(occurrences)**

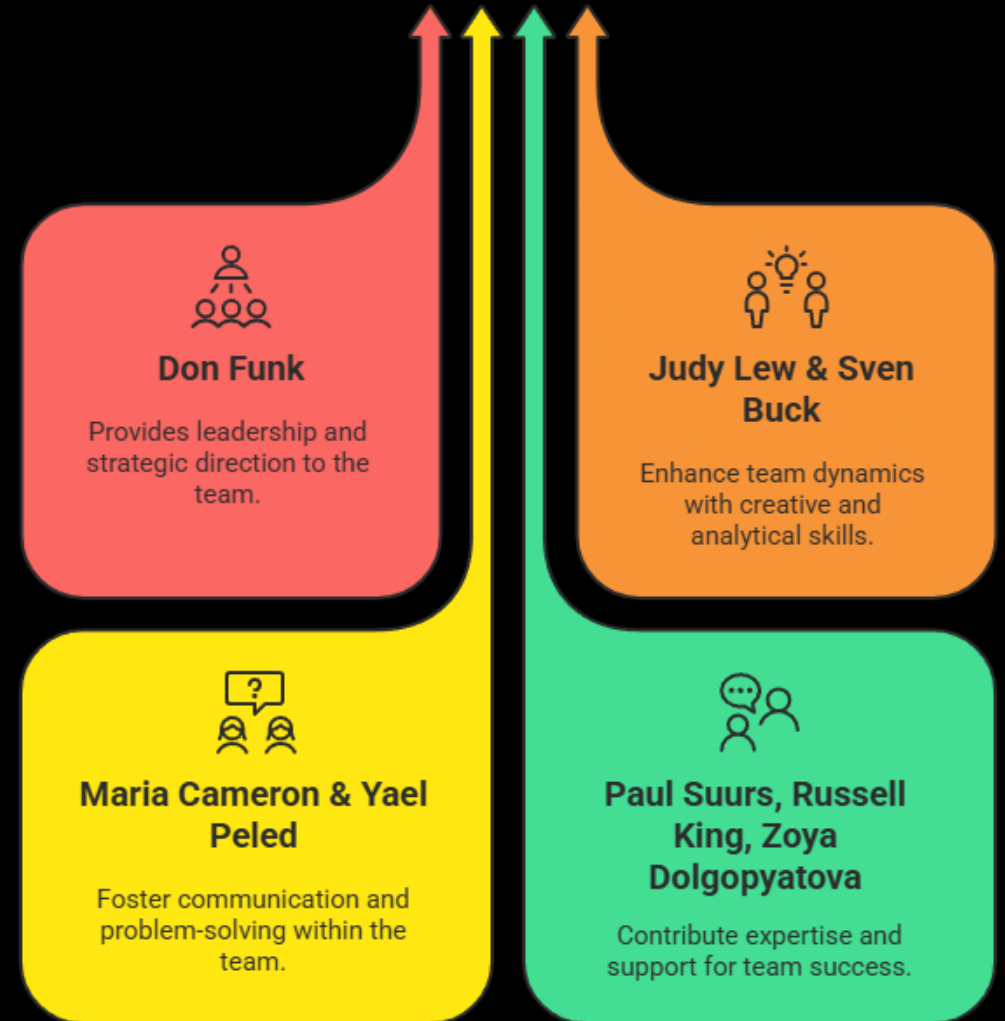


Made with  Napkin

-- WHICH EMPLOYEES HAVE THE SAME MANAGER (MGR ID)?

```
SELECT MGRID ,  
GROUP_CONCAT(CONCAT(FIRSTNAME, ' ',  
LASTNAME) ORDER BY FIRSTNAME  
SEPARATOR ', ') AS Employees  
FROM NORTHWINDCORE.EMPLOYEE  
WHERE MGRID IS NOT NULL  
GROUP BY MGRID ;
```

Building a Collaborative Team



-- WHICH SHIPPING COMPANY (SHIPPER) IS USED THE MOST?

```
select S2.COMPANYNAME , count(*) as Shipper from  
NORTHWINDCORE.SALESORDER S  
join NORTHWINDCORE.SHIPPER S2  
on S.SHIPPERID = S2.ENTITYID  
group by S2.ENTITYID  
order by Shipper desc  
limit 1 ;
```

-- WHAT IS THE AVERAGE SHIPPING TIME FROM ORDER DATE TO REQUIRED DATE?

```
select ROUND( AVG( DATEDIFF( S.REQUIREDDATE ,  
S.ORDERDATE ) ) ,0 ) as average_shipping  
from NORTHWINDCORE.SALESORDER S ;
```

28

-- WHICH COUNTRY RECEIVES THE MOST SHIPMENTS?

```
select S.SHIPCOUNTRY , count(*) as OrderCount from  
NORTHWINDCORE.SALESORDER S  
group by S.SHIPCOUNTRY  
order by OrderCount desc  
limit 1 ;
```



-- WHICH REGIONS HAVE THE MOST DELAYED SHIPMENTS (SHIPPEDDATE VS REQUIREDDATE)?

```
SELECT S.SHIPREGION, MAX(DATEDIFF(S.REQUIREDDATE,  
S.ORDERDATE)) AS MaxDateDiff  
FROM NORTHWINDCORE.SALESORDER S  
GROUP BY S.SHIPREGION  
ORDER BY MaxDateDiff desc  
limit 1 ;
```

-- WHAT IS THE AVERAGE FREIGHT COST PER ORDER?

```
SELECT ROUND( AVG(S.FREIGHT),2) as average_freight_cost
from NORTHWINDCORE.SALESORDER S ;
```

-- WHICH SUPPLIER PROVIDES THE HIGHEST NUMBER OF PRODUCTS?

```
select s.COMPANYNAME ,  
COUNT(PRODUCTNAME) as PRODUCTCOUNT  
from NORTHWINDCORE.PRODUCT P  
join NORTHWINDCORE.SUPPLIER S  
on P.SUPPLIERID = S.ENTITYID  
GROUP by S.ENTITYID  
order by PRODUCTCOUNT desc  
limit 1 ;
```

```
SELECT SUPPLIER_COUNT,  
GROUP_CONCAT(COMPANYNAME ORDER BY  
COMPANYNAME SEPARATOR ', ') AS SUPPLIERS  
FROM (  
SELECT S.COMPANYNAME,  
COUNT(P.PRODUCTNAME) AS SUPPLIER_COUNT  
FROM NORTHWINDCORE.PRODUCT P  
JOIN NORTHWINDCORE.SUPPLIER S  
ON P.SUPPLIERID = S.ENTITYID  
GROUP BY S.COMPANYNAME  
) AS SupplierCounts  
GROUP BY SUPPLIER_COUNT  
ORDER BY SUPPLIER_COUNT DESC;
```

-- WHAT IS THE TOTAL REVENUE GENERATED BY EACH SUPPLIER'S PRODUCTS?

```
select S.COMPANYNAME , SUM(P.UNITPRICE * P.UNITSINSTOCK)
as TOTAL_REVENUE
from NORTHWINDCORE.PRODUCT P
join NORTHWINDCORE.SUPPLIER S
on P.SUPPLIERID = S.ENTITYID
GROUP by S.COMPANYNAME ;
```



Compare revenue generation extremes among suppliers.

-- WHICH SUPPLIER HAS THE MOST OUT-OF-STOCK PRODUCTS?

```
SELECT S.COMPANYNAME , sum( P.UNITSinSTOCK )  
as out_of_stock  
FROM  
NORTHWINDCORE.PRODUCT P  
join NORTHWINDCORE.SUPPLIER S  
on S.ENTITYID = P.SUPPLIERID  
GROUP by P.SUPPLIERID  
order by out_of_stock asc  
limit 1 ;
```

```
SELECT S.COMPANYNAME,  
COUNT(*) AS OUT_OF_STOCK  
FROM NORTHWINDCORE.PRODUCT P  
JOIN NORTHWINDCORE.SUPPLIER S  
ON S.ENTITYID = P.SUPPLIERID  
WHERE P.UNITSinSTOCK = 0  
GROUP BY S.COMPANYNAME  
ORDER BY OUT_OF_STOCK DESC  
LIMIT 1;
```

-- WHICH SUPPLIER OPERATES IN THE MOST REGIONS?

```
SELECT S.REGION,  
COUNT(DISTINCT S.ENTITYID) AS OUT_OF_STOCK  
FROM NORTHWINDCORE.PRODUCT P  
JOIN NORTHWINDCORE.SUPPLIER S  
ON S.ENTITYID = P.SUPPLIERID  
GROUP BY S.REGION  
limit 1 ;
```

-- WHICH SUPPLIERS HAVE THE HIGHEST-REORDER LEVEL PRODUCTS?

```
SELECT S.COMPANYNAME , sum(P.REORDERLEVEL) as  
reorder_Level  
from  
NORTHWINDCORE.PRODUCT P  
join NORTHWINDCORE.SUPPLIER S  
ON S.ENTITYID = P.SUPPLIERID  
GROUP by S.COMPANYNAME  
order by reorder_Level desc  
limit 1 ;
```

-- WHICH MONTH HAS THE HIGHEST NUMBER OF ORDERS?

SELECT

MONTH(S.ORDERDATE) **AS** *orderMonth*,

MAX(**DATE_FORMAT**(S.ORDERDATE, '%M')) **AS** *monthName*,

SUM(O.QUANTITY) **AS** *totalQuantity*

FROM NORTHWINDCORE.SALESORDER S

JOIN NORTHWINDCORE.ORDERDETAIL O

ON O.ORDERID = S.ENTITYID

GROUP BY *orderMonth*

ORDER BY *totalQuantity* **desc**

limit 1 ;

-- WHAT IS THE AVERAGE ORDER VALUE PER CUSTOMER?

```
select C.ENTITYID , C.CONTACTNAME ,  
round(Avg(O.UNITPRICE * O.QUANTITY * (1- O.DISCOUNT) ) ,  
2)as OrderValue  
from NORTHWINDCORE.SALESORDER S  
join NORTHWINDCORE.CUSTOMER C  
on S.CUSTOMERID = C.ENTITYID  
join NORTHWINDCORE.ORDERDETAIL O  
on O.ORDERID = S.ENTITYID  
group by C.ENTITYID  
order by OrderValue desc;
```

This spectrum categorizes customers based on their average order value, ranging from low-value to high-value purchases.



-- HOW MANY ORDERS ARE PLACED IN EACH REGION?

```
select S.SHIPREGION , count(*)  
from NORTHWINDCORE.SALESORDER S  
GROUP by S.SHIPREGION;
```

-- WHAT IS THE MOST FREQUENTLY ORDERED PRODUCT?

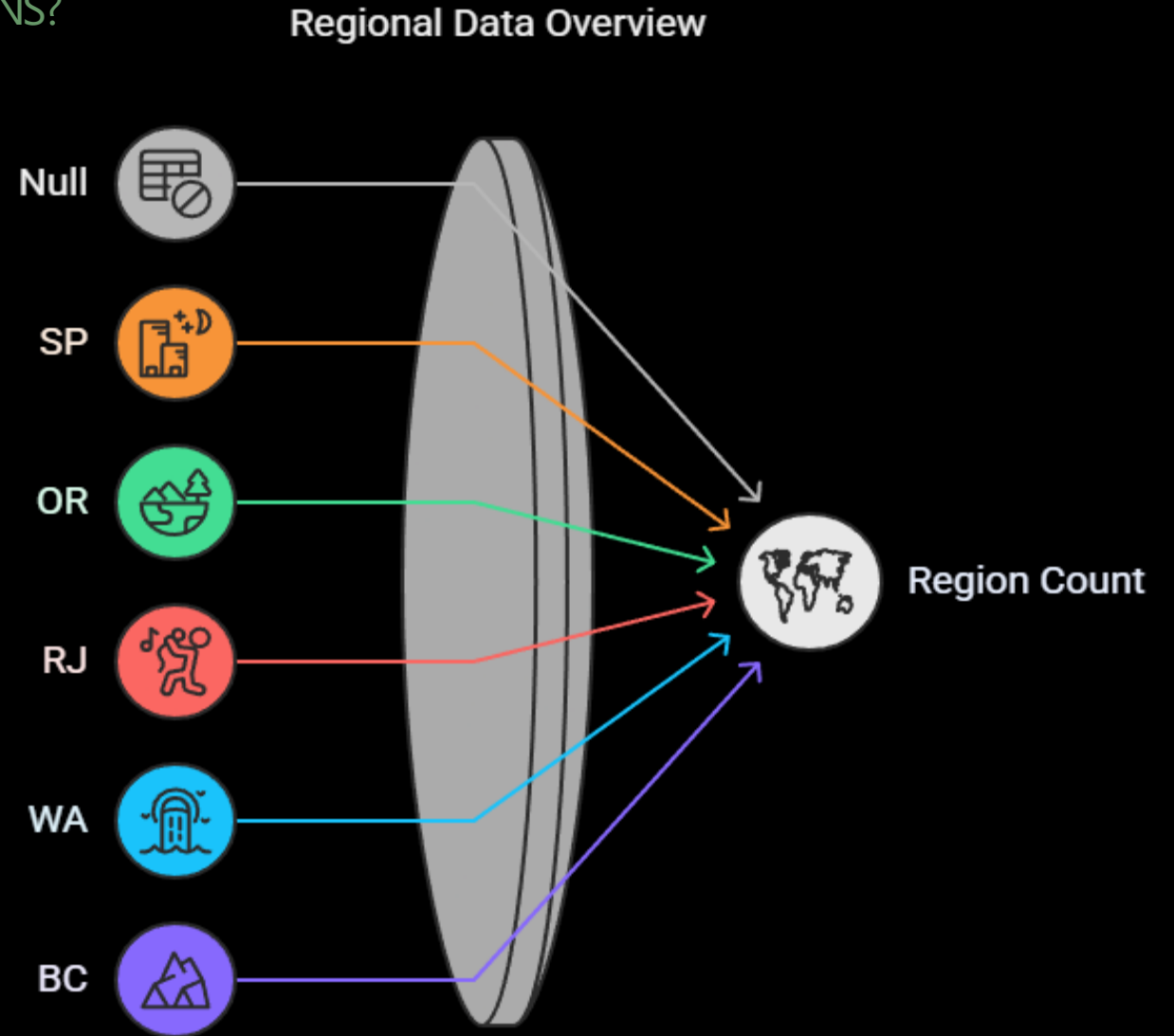
```
select P.PRODUCTNAME , Count(O.ORDERID) as OderCount
from NORTHWINDCORE.ORDERDETAIL O
join NORTHWINDCORE.PRODUCT P
on O.PRODUCTID = P.ENTITYID
group by P.ENTITYID
order by ODERCOUNT desc
limit 1 ;
```

-- WHICH SUPPLIERS HAVE THE HIGHEST-REORDER LEVEL PRODUCTS?

```
SELECT S.COMPANYNAME , sum(P.REORDERLEVEL) as  
reorder_Level  
from  
NORTHWINDCORE.PRODUCT P  
join NORTHWINDCORE.SUPPLIER S  
ON S.ENTITYID = P.SUPPLIERID  
GROUP by S.COMPANYNAME  
order by reorder_Level desc  
limit 1 ;
```

-- WHAT ARE THE MOST COMMON SHIPPING REGIONS?

```
select C.REGION , count(*)  
from NORTHWINDCORE.CUSTOMER C  
group by C.REGION  
order by count(*) desc;
```



-- WHICH PRODUCTS ARE OUT OF STOCK?

```
SELECT P.PRODUCTNAME
FROM NORTHWINDCORE.PRODUCT P
WHERE P.UNITSinSTOCK = 0
UNION
SELECT 'No product is out of stock'
WHERE NOT EXISTS (
SELECT 1
FROM NORTHWINDCORE.PRODUCT
WHERE UNITSinSTOCK = 0
);
```

```
SELECT
CASE
WHEN COUNT(*) = 0 THEN
'No product is out of stock'
ELSE
GROUP_CONCAT(PRODUCT
NAME)
END AS Result
FROM
NORTHWINDCORE.PRODUCT
WHERE UNITSinSTOCK = 0;
```

-- WHAT IS THE AVERAGE QUANTITY PER UNIT FOR EACH CATEGORY?

SELECT

C.CATEGORYNAME,

AVG(CAST(SUBSTRING_INDEX(P.QUANTITYPERUNIT, ' ', 1) AS UNSIGNED)) AS AVG_QUANTITY_PER_UNIT

FROM

NORTHWINDCORE.CATEGORY C

JOIN

NORTHWINDCORE.PRODUCT P

ON

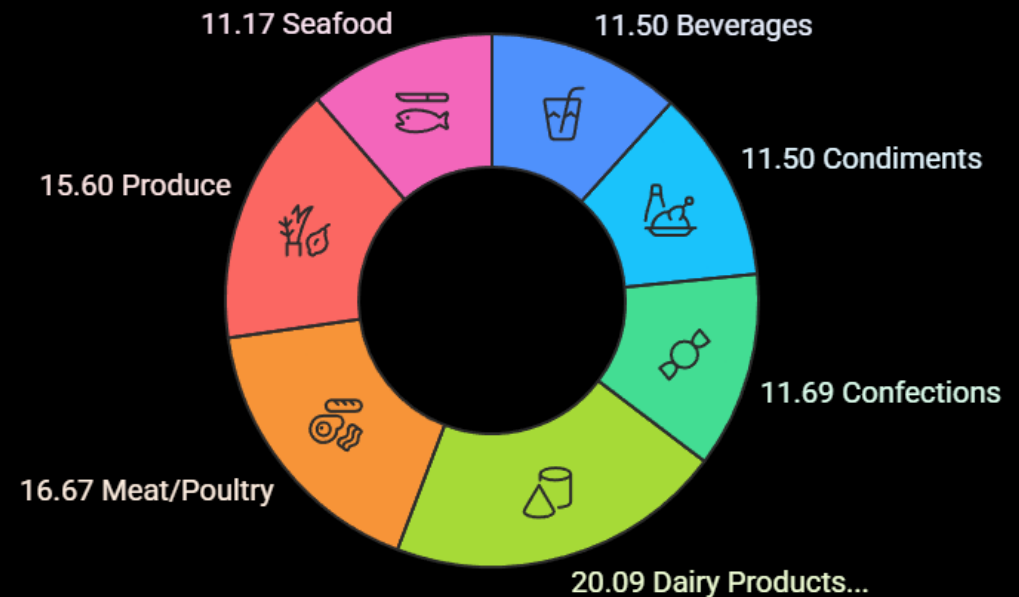
C.ENTITYID = P.CATEGORYID

WHERE

P.QUANTITYPERUNIT IS NOT NULL

GROUP BY

C.CATEGORYNAME;



Made with Napkin

-- WHICH PRODUCTS HAVE THE HIGHEST UNIT PRICE?

```
select *  
from NORTHWINDCORE.PRODUCT P  
where  
P.UNITPRICE = (  
select max(X.UNITPRICE) from NORTHWINDCORE.PRODUCT X  
);
```

-- WHICH REGIONS HAVE THE HIGHEST DEMAND FOR STOCK REPLENISHMENT?

SELECT

S.REGION,

SUM(P.UNITSinSTOCK) **AS** TOTAL_UNITS_IN_STOCK,

SUM(P.UNITSONORDER) **AS** TOTAL_UNITS_ON_ORDER,

(**SUM**(P.UNITSONORDER) - **SUM**(P.UNITSinSTOCK)) **AS** STOCK_DEMAND

FROM

NORTHWINDCORE.PRODUCT P

JOIN

NORTHWINDCORE.SUPPLIER S

ON

P.SUPPLIERID = S.ENTITYID

GROUP BY

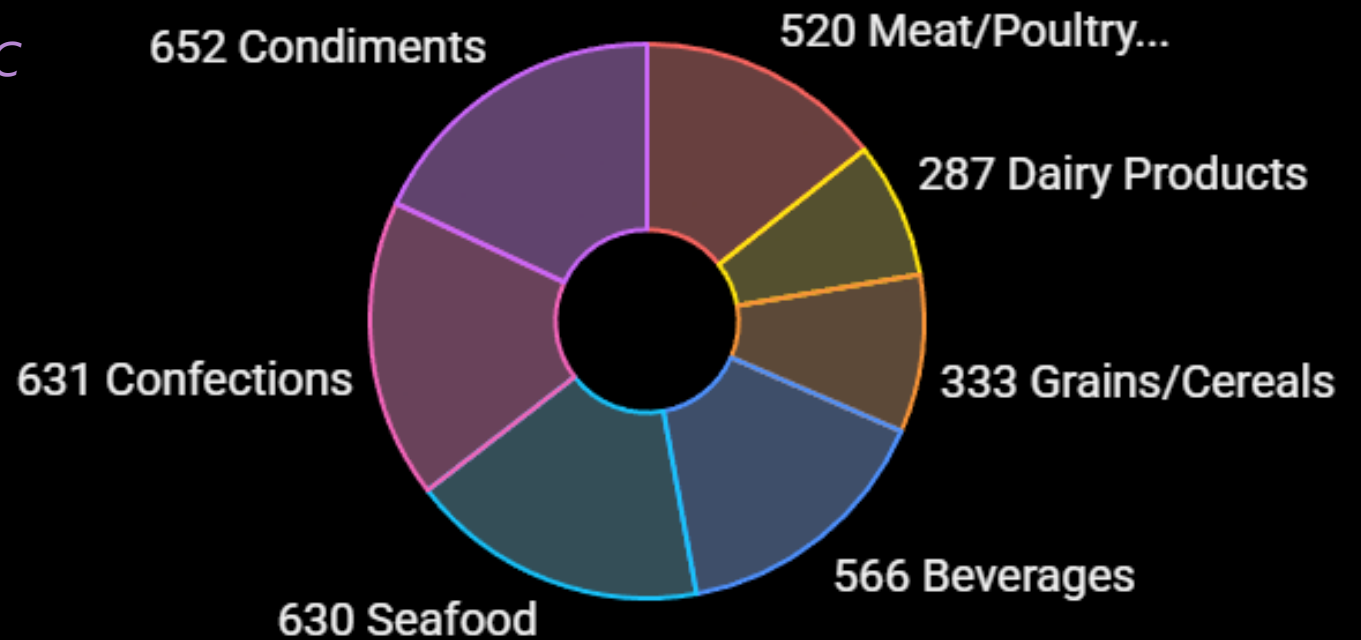
S.REGION

ORDER BY

STOCK_DEMAND **DESC**;

-- WHICH PRODUCT CATEGORIES ARE RUNNING LOW ON INVENTORY?

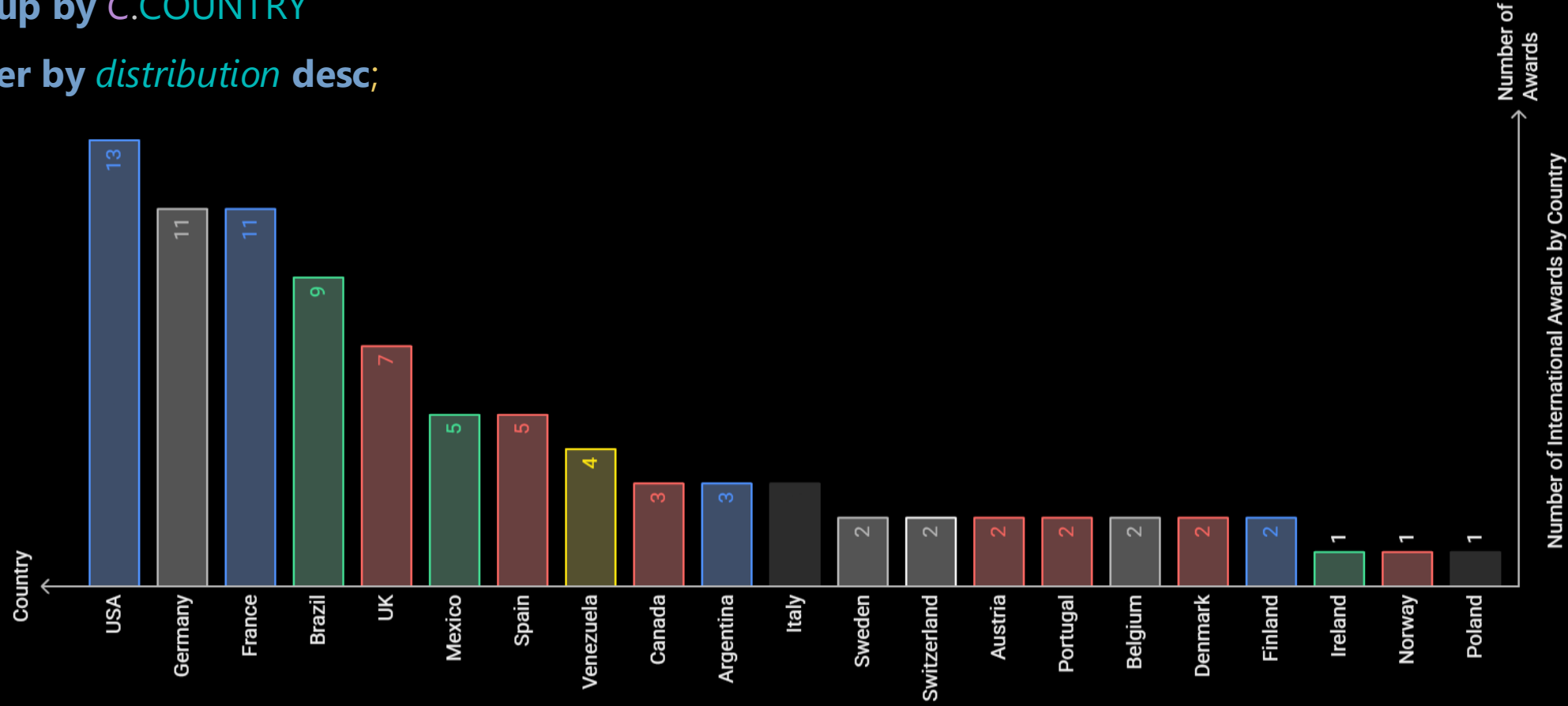
```
select C.CATEGORYNAME , sum(P.UNITSinSTOCK) lowinventory
from
NORTHWINDCORE.PRODUCT P
join NORTHWINDCORE.CATEGORY C
on P.CATEGORYID = C.ENTITYID
group by C.CATEGORYNAME
order by lowinventory ;
```



Made with Napkin

-- WHAT IS THE DISTRIBUTION OF CUSTOMERS BY COUNTRY?

```
select C.COUNTRY , count(*) distribution
from NORTHWINDCORE.CUSTOMER C
group by C.COUNTRY
order by distribution desc;
```



Made with Napkin

```
-- WHAT IS THE AVERAGE NUMBER OF ORDERS PER CUSTOMER?  
SELECT AVG(ORDER_COUNT) AS AVG_ORDERS_PER_CUSTOMER  
FROM (  
  SELECT  
    C.ENTITYID,  
    COUNT(S.ENTITYID) AS ORDER_COUNT  
  FROM  
    NORTHWINDCORE.SALESORDER S  
  JOIN  
    NORTHWINDCORE.CUSTOMER C  
  ON  
    C.ENTITYID = S.CUSTOMERID  
  GROUP BY  
    C.ENTITYID  
) AS CUSTOMER_ORDERS;
```

-- WHICH CUSTOMERS PLACE REPEAT ORDERS THE MOST?

```
select C.COMPANYNAME, count(S.ENTITYID) as ORDER_COUNT
from NORTHWINDCORE.SALESORDER S
join
NORTHWINDCORE.CUSTOMER C
on
S.CUSTOMERID = C.ENTITYID
group by C.ENTITYID
HAVING ORDER_COUNT > 1
ORDER BY ORDER_COUNT DESC;
```

-- WHAT PERCENTAGE OF CUSTOMERS USE MOBILE CONTACT INSTEAD OF LANDLINE?

SELECT

ROUND(

(**COUNT**(**CASE WHEN MOBILE IS NOT NULL AND (PHONE IS
NULL OR PHONE = '') THEN 1 END**)

/ COUNT(*****)) * 100, 2

) AS PERCENTAGE_MOBILE_ONLY

FROM

NORTHWINDCORE.CUSTOMER;

-- WHAT IS THE MOST FREQUENTLY ORDERED PRODUCT CATEGORY?

```

SELECT
C.CATEGORYNAME,
SUM(O.QUANTITY) AS TOTAL_QUANTITY_ORDERED
FROM
NORTHWINDCORE.ORDERDETAIL O
JOIN
NORTHWINDCORE.PRODUCT P ON O.PRODUCTID = P.ENTITYID
JOIN
NORTHWINDCORE.CATEGORY C ON C.ENTITYID = P.CATEGORYID
GROUP BY
C.CATEGORYNAME
ORDER BY
TOTAL_QUANTITY_ORDERED DESC
LIMIT 1;
    
```

Product Category Order Frequency

Category	Order Count
 Beverages	9532
 Dairy Products	9149
 Confections	7906
 Seafood	7681
 Condiments	5298
 Grains/Cereals	4562
 Meat/Poultry	4199
 Produce	2990

-- WHICH REGION GENERATES THE MOST REVENUE ?

SELECT

S.SHIPREGION,

SUM(O.UNITPRICE * O.QUANTITY * (1 - O.DISCOUNT)) **AS** TOTAL_REVENUE

FROM

NORTHWINDCORE.SALESORDER S

JOIN

NORTHWINDCORE.ORDERDETAIL O **ON** S.ENTITYID = O.ORDERID

GROUP BY

S.SHIPREGION

ORDER BY

TOTAL_REVENUE **DESC**

LIMIT 1;

-- WHAT IS THE TOTAL SALES REVENUE PER YEAR?

SELECT

Year(S.ORDERDATE) as *Year_AS* ,

round(**SUM**(O.UNITPRICE * O.QUANTITY * (1 - O.DISCOUNT)) ,2)

AS *TOTAL_REVENUE*

FROM

NORTHWINDCORE.SALESORDER S

JOIN

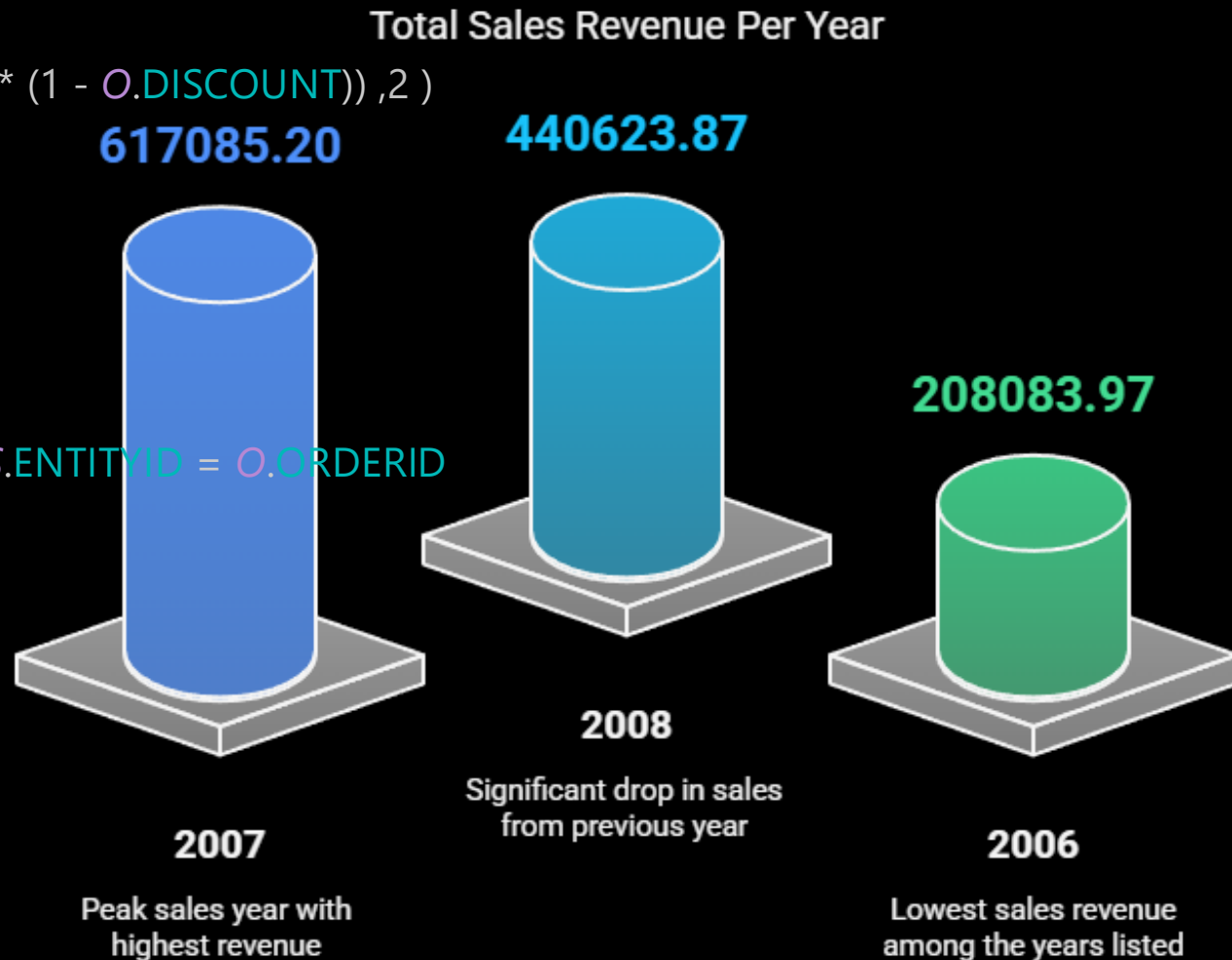
NORTHWINDCORE.ORDERDETAIL O **ON** S.ENTITYID = O.ORDERID

GROUP BY

Year(S.ORDERDATE)

ORDER BY

TOTAL_REVENUE **DESC** ;



-- WHAT IS THE GROWTH TREND IN ORDER VOLUME OVER THE YEARS?

SELECT

YEAR(S.ORDERDATE) **AS** YEAR_AS,

COUNT(S.ENTITYID) **AS** TOTAL_ORDERS,

LAG(**COUNT**(S.ENTITYID), 1) **OVER** (**ORDER BY YEAR**(S.ORDERDATE)) **AS** PREVIOUS_YEAR_ORDERS,

ROUND(

((**COUNT**(S.ENTITYID) - **LAG**(**COUNT**(S.ENTITYID), 1) **OVER** (**ORDER BY YEAR**(S.ORDERDATE)))

/ **NULLIF**(**LAG**(**COUNT**(S.ENTITYID), 1) **OVER** (**ORDER BY YEAR**(S.ORDERDATE)), 0)) * 100, 2

) **AS** GROWTH_PERCENTAGE

FROM

NORTHWINDCORE.SALESORDER S

GROUP BY

YEAR(S.ORDERDATE)

ORDER BY

YEAR_AS;

